TN-March-2015

Class-12 Chemistry

PART III

Time Allowed :3 Hours **Maximum Marks : 150**

Instructions:

1. Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.

2. Use Black or Blue ink to write and pencil to draw diagrams.

Answer all the questions. Choose and write the correct answer

Part – I **(30×1=30)**

(i) Answer all the questions

(ii) Choose and write the correct answer.

1. The number of moles of electrons required to discharge one mole of Al^{+3} is:

(a) 3

(b)1

(c) 2

(d) 4

Ans. (a) 3

-Three moles required as Al3+ lacks 3 electrons.

2. In which of the following processes, the process is always non feasible?

(a) $\Delta H > 0, \Delta S > 0$

(b) $\Delta H<0, \Delta S>0$

(c) $\Delta H > 0, \Delta S < 0$

(d) $\Delta H < 0, \Delta S < 0$

Ans. (c) $\Delta H > 0, \Delta S < 0$

3. Methyl Ketones are usually characterized by:

(a) The Fehling's solution

(b) The iodoformism test

(c) The Schiff's test

(d) The tolan's reagent

Ans. (b) The iodoform test

- When it is heated with iodine in presence of sodium hydroxide yellow ppt of Iodoform is obtained.

4. The acid that cannot be prepared by Grignard reagent is:

(a) Acetic acid

(c) Butyric acid

(b) Formic acid

(d) Benzoic acid

Ans. (b) Formic acid

5. In case of physical adsorption there is desorption when:

(a) Temperature increases

(c) Pressure increases

(b) temperature decreases

(d) concentration increases

Ans. (a) Temperature increases

- (Physical absorption is a reversible process.)

6. Oxidation of aniline with acidified potassium dichromate gives:

(a) p-benzoquinone

(c) Benzaldehyde

(b) benzoic acid

(d) benzyl alcohol

Ans. p – bezoquinone

7. An emulsion is a colloidal solution of:

(a) Two solids

(c) Two gases

(b) Two liquids

(d) One solid and one liquid

Ans. (b) Two liquids

8. Which one of the following does not form peroxide easily?

(a) Diethyl ether

(b) Ethyl methyl ether

(c) Dimethyl ether

(d)Anisole

Ans. (d) Anisole

9. Which compound is used as soil sterilizing agent?

(a) Nitrobenzene

(b) Nitromethane

(c) Aniline

(d) Chloropicrin

Ans. (d) Chloropicrin

10. In the nuclear reaction, ${}_{97}U^{238} \rightarrow {}_{82}Pb^{208}$, the number of α and β particles emitted are:

(a) 7 α, 5β

(b) 6 α, 4β

(c) 4 α, 3β

(d) 8 α, 6β

Ans. (d) 8 α, 6β

11. For the homogeneous gas reaction at 600 K $4NH_3\ (_g) + 5O_2(_g) \rightleftharpoons 4NO\ (_8) + 6H_2O(_g)$, the equilibrium constant K_C, has the unit:

(a) $(mol\ dm^{-3})^{-1}$

(b) $(mol\ dm^{-3})$

(c) $(mol\ dm^{-3})^{11\circ}$

(d) $(rnoldm^{3})$ `9

Ans. (b) $(mol\ dm^{-3})$

12. The actinide contraction is clue to:

(a) Perfect shielding of 5f electron

(b) Imperfect shielding of 4f electron

(c) Imperfect shielding of 5f electron

(d) Perfect shielding of 4f electron

Ans. (c) Imperfect shielding of 5f electron

13. Glucose is not oxi4ised to gluconic acid by:

(a) $\frac{Br_2}{H_2O}$

(b) Fehling's solution

(c) Tollens's reagent

(d) Conc. HNO_3

Ans. (a) $\frac{Br_2}{H_2O}$

14. Which molecule is relatively more stable?

(a) O_2

(b) H_2

(c) Li_2

(d) N_2

Ans. (d) N_2

15. When phenol is distilled with Zn - dust, it gives:

(a) benzaldehyde

(b) benzoic acid

(c) toluene

(d) benzene

Ans. (d) benzene

16. The principal emulsifying agent for W/O emulsion is:

(a) Proteins

(b) Gum

(c) Lamp 'black

(d) Synthetic soaps

Ans. (b) Gum

17. Noble gases have electron affinity.

(a) High

(b) Low

(c) Zero

(d) Very low

Ans. (c) Zero

-Since, they already have completely filled valance shell.

18. The shape of XeF_4 is:

(a) Tetrahedral

(b) Octahedral

(c) Square planar

(d)Pyramidal

Ans. (c) Square planer

19. The alloy used in the manufacture of resistance wires is:

(a) Ferro - chrome

(b) Bronze

(c) Nichrome

(d) Satellite

Ans. (c) Nichrome

- Since they have very high resistance.

20. Which one of the following is a polysaccharide?

(a) Sucrose
(b) Cellulose
(c) Maltose
(d) Raffinose

Ans. (b) Cellulose

21. The total number of atoms per unit cell of bcc is:

(a) 1
(b) 2
(c) 3
(d) 4

Ans. (b) 2

22. the change in entropy for a system and surrounding are 0.228)1C-' and +0.260 JK^{-1} respectively. Then entropy change of the universe is:

(a) -0.0313 JK^{-1}
(b) +0.0313 JK^{-1}
(c) +0.877 jK^{-1}
(d) -0.877 JK^{-1}

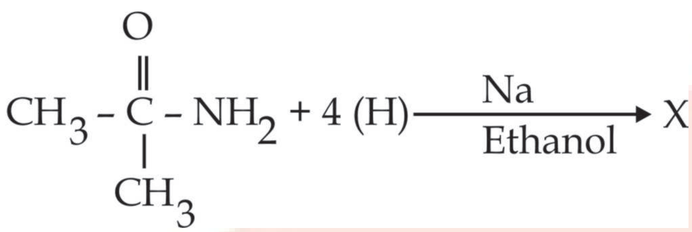
Ans. (b) +0.0313 JK^{-1}

23. The function of ferredoxin is:

(a) photosynthesis
(b) storage and transport of oxygen
(c) electron transfer
(d) sensitizer

Ans. (a) photosynthesis

24.



The component X is:

(a) Methylamine

(b) Ethylamine

(c) Dimethylamine

(d) Nitro - methane

Ans. (b) Ethylamine

25.__________ is used in gas lamp material.

(a) MnO_2

(b) CeO_2

(c) N_2O_5

(d) Fe_2O_3

Ans. (c) N_2O_5

26.The value of Bohr radius for hydrogen atom is:

(a) 0.529×10^{-8} cm

(b) 0.529×10^{-10} cm

(c) 0.529×10^{-6} cm

(d) 0.529×10^{-12} cm

Ans. (d) 0.529×10^{-12}cm

27. Which of the following compounds will not give positive chromyl chloride test?

(a) $CuCl_2$

(b) $HgCl_2$

(c) $ZnCl_2$ (d) C_6H_5Cl

Ans. (b) $HgCl_2$

28. Oxygen atom of ether is:

(a) very reactive (c) oxidizing

(b) replaceable (d) comparatively inert

Ans. (d) comparatively inert

29. The excess energy which a molecule must possess to become active is known as:

(a) Kinetic energy (c) Potential energy

(b) Threshold energy (d) Activation energy

Ans. (d) Activation energy

30. For an equilibrium reaction, Ling value is positive, which of the following is correct?

(a) $K_P = K_c$ (c) $K_P < K_c$

(b) $K_P > K_c$ (d) $K_p = K_c = 0$

Ans. (b) $K_P > K_c$

Part – II (15×3=45)

(i) Answer any fifteen questions.

(ii) Each answer should be in one or two sentences.

31. Define hybridization.

Ans. Hybridization is process of combination of two or more orbitals of same energy level.

32. The experimental value of d (Si — c) is 1.93 **Å**. If the radius of carbon is 0.77 **Å**, calculate the radius of silicon.

Ans. The experimental value of d(Si-C) is 1.93Ao. Thus,

r(Si) = d (Si - C) - r(C)

= 1.93 - r(C)

= 1.93 - 0.77[$\because$ r(C) = 0.77Ao] = 1.16Ao

33. What are the inter halogen compounds? Give one preparation.

Ans. An interhalogen compound is a molecule which contains two or more different halogen atoms and no atoms of elements from any other groups.

34. H_3PO_4 is triprotic, justify with equations.

Ans. $H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$

$H_3PO_4 + 2NaOH \rightarrow Na_2HPO_4 + 2H_2O$

$$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$$

35. Why do d-block elements form complexes?

Ans. Availability of Vacant d- orbital's –Transitions elements have vacant d- orbital's electrons confine ns2(n-1) d 0 -10 have empty d orbitals to accommodate as accept electrons coming from ligands that are attached to them.

36. What is spitting of silver? How can it be prevented?

Ans. When molten silver includes the oxygen of the atmosphere, absorbing 20 times its own volume of gas, the oxygen, is not permanently retained for a cooling it is expelled with great violence this phenomenon is known as spitting of silver. This can be prevented by covering the molten metal with a layer of charcoal.

37. Write the principle behind the Hydrogen bomb.

Ans. The principle behind hydrogen bomb is a process called nuclear fusion had been observed by physics that a fusion of nucleus takes place in either amount of heat is produced in the form of heat and light.

38. How are glasses formed?

Ans. Glass can be made by heating ordinary sand (which is mostly mad of silicon dioxide) until melts and turns into liquid.

39. What is entropy? Write its units.

Ans. It is thermodynamic function that describes the randomness and disorder of molecules based on the number of different arrangements available to them in a given system of reaction. Its SI unit of (J) per molar Kelvin (Jmol-1k-1).

40. State Le - chatline's principle.

Ans. When a change in concentration pressure is temperature is applied to system at equilibrium the position of the equilibrium shifts in the direction that tends to reduce the effect of the change.

41. What are simple and complex reactions?

Ans. Simple reaction is elementary reaction or is separated. In some reaction may side reaction occur along with the main reaction including product formation reaction which does not occur in a single step and takes a sequence of number of elementary reactions are called complex reaction.

42. The rate constant of a first order reaction is 1.54 x 10^{-3}sec $^{-1}$. Calculate its half-life period.

Ans. Given: rate constant of first order reaction= 1.54× 10^-3 sec-1.

$$\frac{t1}{2} = \frac{0.693}{k}$$

$$\frac{t1}{2} = \frac{0.693}{1.54 \times 10^{-}3sec - 1}$$

$$= 450\,\text{sec}.$$

43. What is catalysis? Give an example.

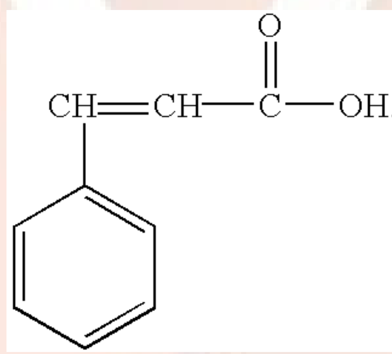
Ans. Catalysis is the process of increasing the rate of a reaction of a chemical reaction by adding a substance known as catalysts which is not consumed in catalysis reaction and can be seen in the product. E.g.: $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$

44. State Ostwald's dilution law.

Ans. Wilhelm Ostwald's detection law is a relationship proposed between the dissolution constant KD and degree of dissociation α of a weak electrolyte.

45. Give the structures of Z and E forms of cinnamon acid.

Ans. Cinnamic acid is 3-phenylpropenoic acid.



Each alkene carbon has two different groups, so the molecule can exist as E/Z isomers

46. Give two chemical tests to distinguish propane - 2 - ol and 2-methylpropan - 2 - ol.

Ans.

1. Lucas test $\rightarrow$ propane -2-a; testing in 2-5 min.

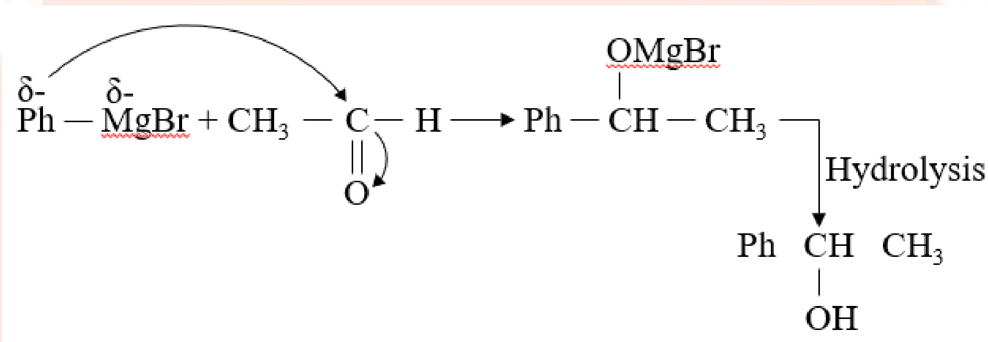
2 methyl propan-2-ol inert tertidity.

2. Victor Meyer test $\rightarrow$ propan-2ol-given Blue Colour.

2 methyl propan-2-ol remain colorless.

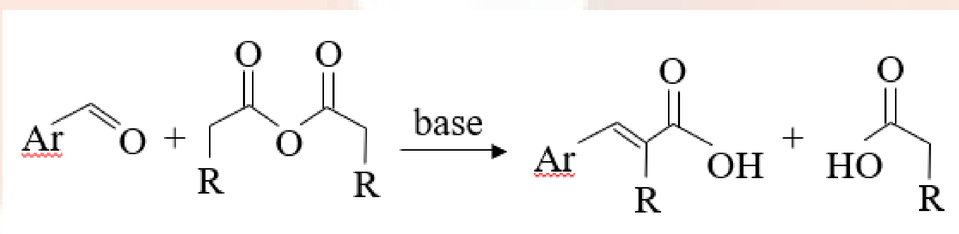
47. How is benzyl alcohol obtained from phenyl magnesium bromide?

Ans.



48. Write Perkins reaction.

Ans. Perkins Reaction



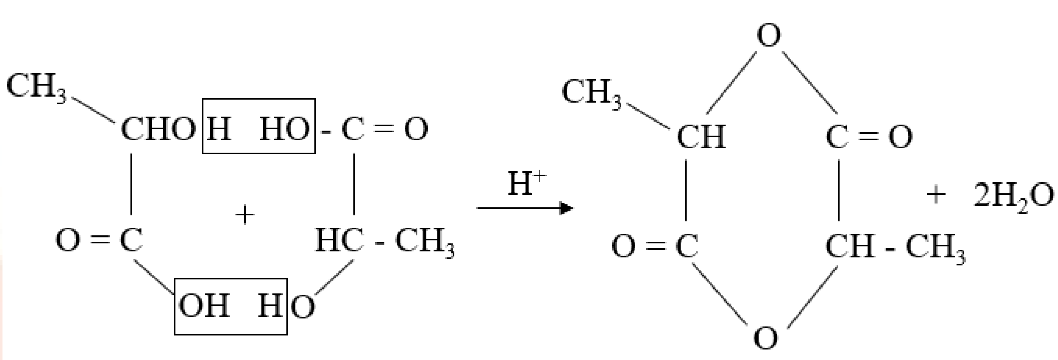
49. Give the structures of chloride and lactide.

Ans.

Chloride:

$$CH_3CH(OH)COOH + PCl_5 \rightarrow \begin{array}{l} CH_3-CH-COCl \\ \qquad\quad | \\ \qquad\quad CL \end{array}$$

Lactide:



50. $C_3H_5CONH_2 \xrightarrow[KOH]{Br_2} X \xrightarrow[HCl]{HNO_2} Y \xrightarrow[Cu^+]{NaNO_2} Z$

Identify X, Y and Z.

Ans. $X = CH_2NH_2$

$Y = CH_2N_2Cl$

$Z = CH_2Cl$

$$C_3H_5CONH_2 \xrightarrow[KOH]{Br_2} CH_2NH_2 \xrightarrow[HCl]{HNO_2} CH_2N_2Cl \xrightarrow[Cu^+]{NaNO_2} CH_2Cl$$

51. What are food preservatives? Give an example.

Ans. Food preservatives are chemicals that prevent food from spoilage due to microbial growth. Table salt, vegetable oil, sodium benzoate (C_6H_3COONa), salts of propanion acid is the examples.

Part III (7×5=35)

Answer any seven questions choosing any two from each section.

SECTION – A

52. Explain the postulates of Molecular orbital theory.

Ans. The atomic orbital's combine to form a new orbital known as molecular orbitals. It loses their identity. Its energy states of molecule. It given electron probability.

53. How is gold extracted from its ore?

Ans. Most gold ore now come from either open pit or underground mines, in all methods of gold are refining the ore is usually washed a filtered at the mine, then sent to the mile. The ore is ground into smaller particles with water, then ground again in a ball mill to further pulverize ore.

54. Write the uses of Lanthanides.

Ans. Lanthanides have been widely used as alloys to impart strength and hardness to metals. The main lanthanide used for their purpose is cerium, mixed with small amounts of lanthanum, neodymium, and praseodymium also use in petroleum industry.

55. Explain coordination and ionization isomerism with examples.

Ans. Ionization isomers are identical except for a ligand has exchanged places with an anion or neutral molecule that was originally outside the coordination complex.

Ex- an octahedral isomer will have five ligands 5 – identical but 6- will differ.

SECTION – B

56. Give the characteristics of free energy.

Ans. Free energy 'G' G is defined as (H-TS), H and S are the enthalpy and entropy of system. T = temperature, DG<O (negative) process is spontaneous and feasible DG = O.

57. Derive the expressions K_C and K_P for decomposition of PCI_5.

Ans. Quantitative Effect of Volume on Equilibrium Position

PCl_5 dissolves to give PCl_3 and Cl_2

$$PCl_3(g) \rightleftharpoons (g) + Cl_2(g)$$

$$K_\varepsilon = \frac{X^2}{V(a-X)}$$

The dissociation of N_2O_4 gives NO_2 gas

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$$K_\varepsilon = \frac{4X^2}{V(a-X)}$$

58. Write the characteristics of first order reaction.

Ans. This implies that the role constant does not depend on the concentration of the reactant. The time taken for half reactant to get reacted, the half life time of 1st order reaction is independent of the concentration.

59. Calculate the e.m.f of the Zn-Ag cell at 25°C when $[Zn^{2+}]$ = 0.001M and $[Ag^{+}]$ = 0.1M. (E^{o}_{cell} at 25°C = 1.56V)

Ans. The cell reaction in the zinc - silver cell would be:

$2Ag^{+} + Zn \rightarrow 2Ag + Zn^{2+}$

The Nernst equation for the above cell reaction may be written as:

$$\frac{E_{cell} = E_{ocell} - RT}{nFln\left[Ag\right]_2\left[Zn^{2+}\right] \ / \ \left[Ag^{+}\right]_2\left[Zn\right]}$$

(Since concentrations of solids are taken as unity i.e., [M] = 1)

$$\frac{E_{cell} = E_{ocell} - RT}{nFln\left[Zn^{2+}\right] \ / \ \left[Ag^{+}\right]_2}$$

Substituting the various values in Nernst equation, we have

$$E_{cell} = \frac{1.56 \ - \ (2.303 \ \times \ 8.314 \ \times \ 298}{\frac{2 \ \times \ 96495) \ log \ 0.1}{102}}$$

$= 1.56 - 0.02957 \, log \, 10 - 3$

$= 1.56 + 3(0.02957) = 1.56 + 0.0887$

$= 1.6487 \, volts$

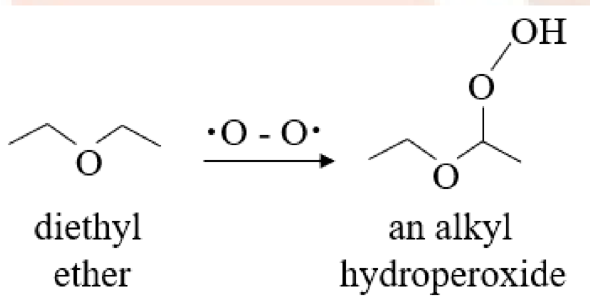
SECTION – C

60. How does diethyl ether reacts with the following reagents?

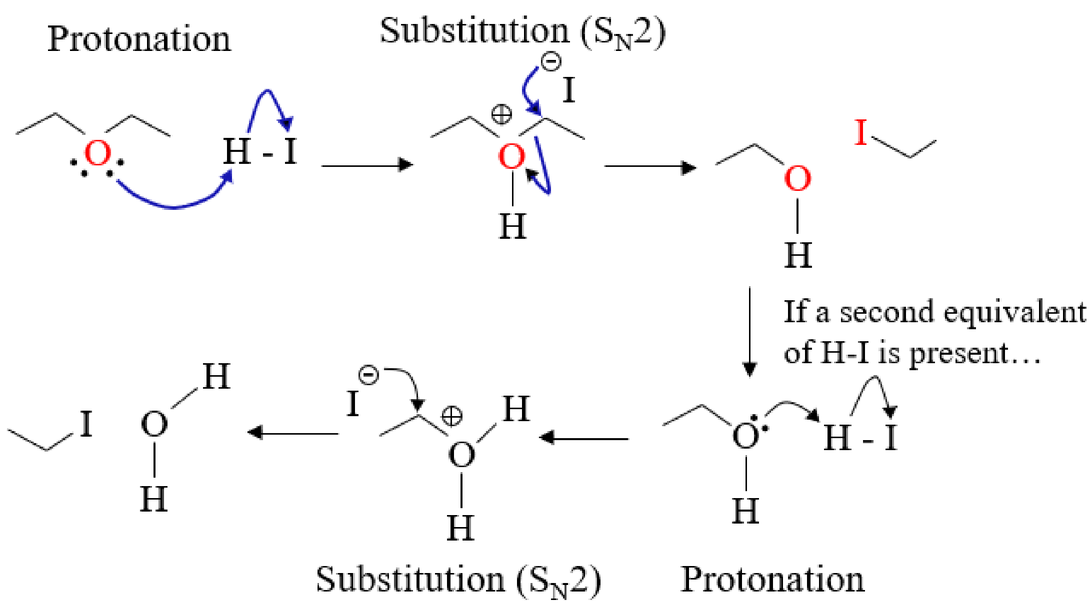
(a) O_2/long contact (b) HI in excess (c) PCI5

Ans.

(a)



(b)

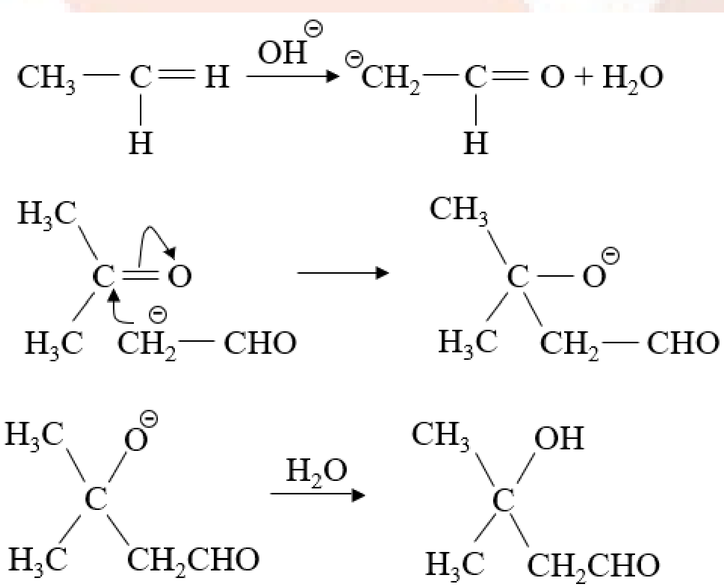


(c) $C_2H_5OC_2H_5 + 2PCL_5 \rightarrow 2C_2H_5CL + 2POCL_3 + HCL$

Dicthyeethor Ethyle Chloride Phosphoryl Chloride

61. Explain the mechanism of acetaldehyde involved crossed aldol condensation.

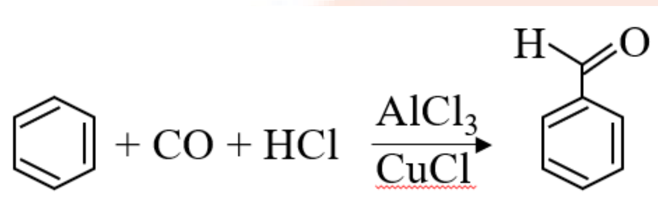
Ans.

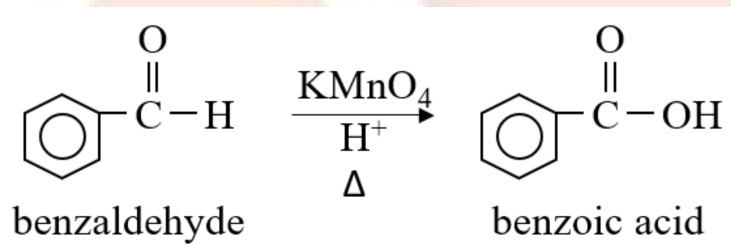


62. How are the following compounds obtained from benzoic acid?

a. Ethyl benzoate b. Benzyl alcoholc. Benzene

Ans.





63. Write briefly on antibiotics. In what way antispasmodics are helpful?

Ans. An antibiotic is a type of antimicrobial substance active against bacteria and is most bacterial infection. Antispasmodics agent is used for smooth muscle relaxation, is tubular organs of gastrointestinal tract.

Part – IV **(4×10=40)**

(i) Answer four questions in all

(ii) Question number 70 is compulsory and answers any three from the remaining questions.

64. A. Explain Pauling's scale of electronegativity.

B. How is fluorine isolated from fluorides?

Ans. A. Pauling scale is a numerical scale of electronegativity based on bond energy calculation for different element joined by covalent bonds.

B. New element and named it fluorine from Ampere the isolating fluorine by electrolysis of dry potassium hydrogen fluoride (KHF2) and dry hydro fluorine acid.

65. $[Co\ (NH_3)_4Cl_2]\ NO_2$

a. IUPAC name

b. Central metal ion

c. Ligandd

d. Charge on the co-ordination sphere

Ans. (a) Tetra ammine chloride nitrito-N-cobalt (III) ion

66. A. Write the characteristics of ionic crystals.

B. How can you determine the charge of the salt particles?

Ans. A. Ionic crystals have hard high melting points and are brittle. When they melt, the resulting liquids conduct electricity well. These proportions reflect the strong attractive forces between ions of opposite change as well as the repulsion that occur when ions of like charge are placed near each other.

B. Estimate the surface change according the PH of solution.

67. a. Explain Ostwald's dilution law.

b. Discuss the relation between free energy and EMF.

Ans. A. Ostwald's dilution law is the application of law of mass action to weak electrolytes in solution. The constant temperature degree of dissociation of weak electrolyte is directly proportional to square root of if dilution.

B. Gibbs free energy and emf of the cell this reversible work done by galvanic cell is related to decrease in Gibb's energy. When concentration of all react species E (cell) = Eo(cell).

68. a. Differentiate between Enantiomer and Diastereomer.

b. Explain the following reactions:

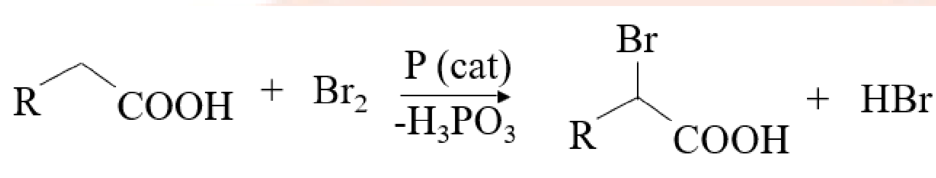
(i) HVZ reaction

(ii) Trans esterification's

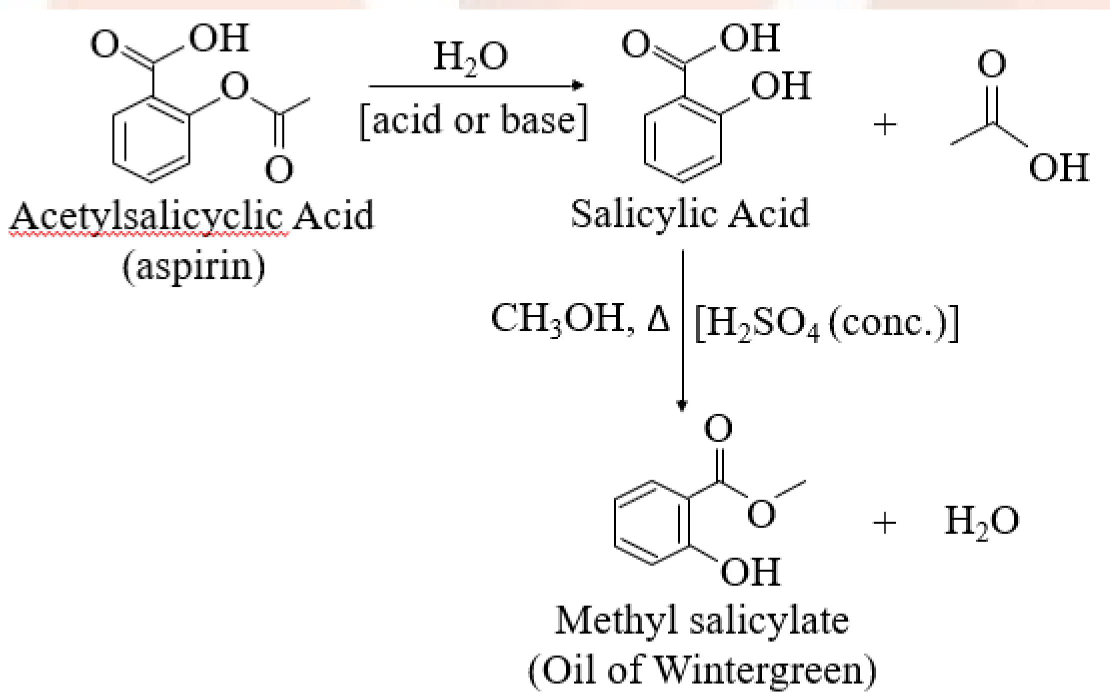
(iii) Methyl salicylate formation

Ans. a) There are two types of stereoisomers - enantiomers and diastereomers. Enantiomers contain chiral centers that are mirror images and non-superimposable. Diastereomers contain chiral centers that are non-superimposable but are NOT mirror images. There can be many more than 2 depending on the number of stereocenters.

b) (i) HVZ reaction



(ii) Trans esterification's



69. a. Distinguish between Primary, Secondary and Tertiary amines.

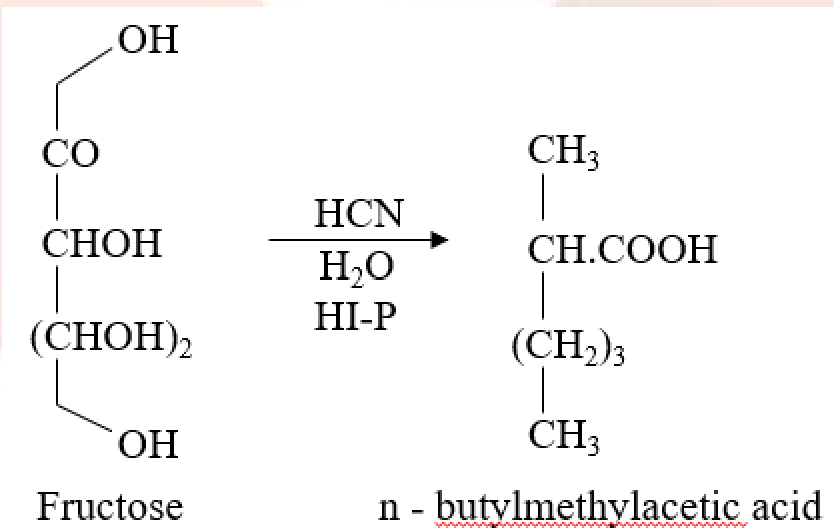
b. Elucidate the structure of fructose.

Ans. a) The main difference between primary secondary and tertiary amines is that, in primary amines, one alkyl or aryl group is attached to the nitrogen atom and in secondary amines, two alkyl or aryl groups are attached to the nitrogen atom whereas, in tertiary amines, three alkyl or aryl groups are attached to the nitrogen

b) Fructose on treatment with hydrogen cyanide gives cyanohydrin which on hydrolysis followed by reduction hydridic acid and red phosphorus give n - butylmethylacetic acid, $CH_3CH_2CH_2CH_2CH\ (CH_3).COOH$

The formation of this compound again indicates that the ketonic group present at second position.

The straight chain structure of fructose may be written as



70. (a) An organic compound (A) of molecular formula C_2H_4 reacts with alkaline potassium permanganate and gives compound (B) of molecular formula $C_2H_6O_2$. Compound (B)

when heated with anhydrous zinc chloride forms (C) of molecular formula C_2H_4O. Identify A, B and C and explain the reactions.

(b)An element (A) belongs to group number 12 and period number 4, bluish white in color reacts with Conc.H_2SO_4 to give its salt (B) with the liberation of SO_2 gases. Compound (B) reacts with sodium bicarbonate to give (C) which is used as ointment for curing skin diseases. Identify A, B and C and explain the reactions.

OR

(a) An organic compound (A) of molecular formula C_7H_8 on oxidation with air and in presence of V_2O_5 to form (B) of molecular formula C_7H_6O. (B) On reduction with lithium aluminum hydride to - form (C) of molecular formula C_7H_8O. Identify (A), (B) and (C) and explain the reactions.

(b) Equivalent conductivity of acetic acid at infinite dilution is 390.7 mho cm^2 gm. $equiv^{-1}$ and for 0.1 M acetic acid is 5.2 mho cm^2gm.$equiv^{-1}$. Calculate degree of dissociation, H+ ion concentration and dissociation constant of the acid.

Ans. (a) Benzaldehyde reduces to benzyl alcohol by NaBH4 or LiAlH4

A is C_7H_8 - $C_6H_5CH_3$ toluene

B is C_7H_6O - C_6H_5CHO Benzaldehyde

C is C_7H_8O - $C_6H_5CH_2OH$ Benzylalcohol

(b) K=α2C/1-α

= (0.0133) × (0.0133) ×0.1 / (1-0.133)

= 2.38×10^{-5}M

Solution: α = 1.33

H^+ = 0.00133M

K = 2.38×10^{-5}M